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A Hybrid Intelligence Framework for Emotion-Aware Deepfake Detection and Misinformation Risk Reduction

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Abstract: The development of deepfakes has accelerated to increase the rates of misinformation, emotional control, and loss of digital trust. The current detection systems mainly aim at seeing or hearing something incongruent and overlook the emotional aspect that compounds psychological and social damage. To fill this gap, a Hybrid Intelligence Framework of Emotion-Aware Deepfake Detection and Misinformation Risk Reduction based on deep learning, affective computing, and explainable AI is proposed in this study. The framework is a combination of spectral spatial forensic, linguistic sentiment, and physiological emotion characteristics via a transformer-directed multimodal fusion network. The hybrid CNN - BiLSTM framework structures spatial forgery localizations and emotional pathway dynamics, which are optimized through adaptive gradient control and attention-guided calibration. The experimental performance on multimodal deepfake datasets shows that the system detects correctly 96.4% with 0.97 AUC and a false emotional trigger rate is reduced by 21 percent over the state-of-the-art systems, which were correspondingly statistically significant ($p < 0.05$). The proposed model increases interpretability, stronghold, and ethical online management through the detection of emotionally manipulative synthetic media and the measurement of the risk of misinformation, which provides a starting point to acquire emotion-stable and reliable AI systems.

Keywords: Deepfake Detection, Emotion-Aware AI, Hybrid Intelligence, Misinformation Risk, Explainable Deep Learning, Transformer Fusion

I. INTRODUCTION

The development of generative artificial intelligence has accelerated the creation of so-called deepfakes, visual or audio media that can convincingly mimic a real person. Such manipulations endanger the integrity of digital space, privacy, and confidence of the population because they disseminate the fake information and create an uncontrolled emotional response. Deepfakes are being employed to trigger anger, fear, or empathy and increase the psychological and societal effect of fake information [1], [2]. The traditional deepfake detectors are based primarily on forensic inconsistency, including pixel anomalies, frequency deformations, or time anomalies, but they do not address the emotional manipulation dimension, which constrains their ability to minimize the risks of misinformation [3], [4]. The recent works have investigated deepfake detection with the help of deep learning models like optimized VGG16, ResNeXt, and hierarchical spectral-feature fusion networks with stunning detection accuracy yet low interpretability and generalization [1], [4], [5]. Transformer-based models have enhanced temporal reasoning and cross-modal representation learning [5], but still, issues of robustness to compression, flexibility to novel generative models, and explainable results are present [6]. Simultaneously, emotion-conscious computing and affective analysis have also proven to be able to estimate emotional tone in digital materials [7]. Nevertheless, the combination of these methods and deepfake forensics is rarely achieved, which creates a big gap between the accuracy of detection and the comprehension of emotional manipulation.

There are special threats of emotion-based misinformation: fake videos and audios that are emotionally colored are agreed upon more quickly and with greater efficiency than real information [8]. Taking into consideration the above manipulations of emotions, such as those identified as being of the wild emotional type, leads to the need to take an integrated approach to the issue, which would involve forensic intelligence (to verify the authenticity) and affective intelligence (to assess the emotional impact). Although the problem is starting to be better acknowledged, few studies are covering the dual problem of identifying the presence of synthetic content and at the same time reducing

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its emotional impact. To fill these gaps, this paper suggests that a Hybrid Intelligence Framework of Emotion-Aware Deepfake Detection and Misinformation Risk Reduction. The framework brings together spectral-spatial forensic attributes, linguistic sentiment representatives, and physiological emotion indicators by incorporating a transformer-mediated multimodal fusion framework. The hybrid model leads to the combination of the convolutional neural networks (CNNs) that perform the localization of spatial forgery with long short-term memory (BiLSTM) modules that model the temporal trajectory of emotions. Attention-based calibration and adaptive gradient regulation is used to improve feature robustness given post-processing conditions. Furthermore, explainable AI (XAI) algorithms like LIME and Grad-CAM present interpretable visual representations to aid forensic reliability [1], [9].

1. A hybrid multimodal deepfake detector combining both the forensic and emotional insights to analyze it thoroughly.
2. A compression-resistant adversarial module able to preserve accuracy with different encoding conditions as well as noise.
3. A justifiable inferential system that generates outputs that can be interpreted by humans regarding digital forensics and misinformation control.An experimental protocol has been proven to have better generalization at zero-day and fewer rates of emotional misclassification.

II. RELATED WORK

Further development of deepfake detection has not been exempted; it was instead the shift in perspective from traditional forensic methods to a more advanced hybrid intelligence framework that incorporates visual, audio, and textual information. The existing literature can be divided into three major directions, namely, forensic-based detection, multimodal transformer frameworks, and emotion-aware explainable systems.

Classical approaches take advantage of the space, frequency, or time differences to detect synthetic materials. [1] created a VGG16 model, optimized and augmented with LIME to predict explainable visual localization of forgery, whereas [4] utilized reconstruction compactness to distinguish between genuine and manipulated samples. The spectral-based models, including HSFF-NET by [5], enhanced detection and localization with the help of hierarchical spectral-feature fusion. These models have high accuracy but have low generalization in the presence of post-processing or compression artifacts.

Recent works use the multimodal fusion and transformer architecture to learn the cross-domain correlations. [9] applied the gaze pattern and blink pattern to the real-time detection of the deepfakes and proved the usefulness of the physiological features. [3] add caption-based functionality to the perceptual quality metrics to make them stronger. [6] have summarized new transformer-based frameworks, which have better generalization. Although these improvements have been achieved, current methods are mostly computationally demanding and do not investigate detecting emotional manipulation and dissemination of misinformation, but rather, they look at authenticity.

Feeling-sensitive computing has been popularized to improve the transparency of AI and learn the behavior of people in relation to digital media. [7] highlighted that explainable multimodal detection frameworks were necessary, whereas Ouyang et al. [10] proposed adaptive perturbation networks on deepfakes of audio. The article by Romero-Moreno [8] and [2] addressed the ethical and psychological aspects of deepfakes; he/she stated that emotion manipulation represents a significant threat. Nevertheless, the existing literature does not involve forensic detection and emotion modeling as seldom as possible to reduce the dissemination of false information.

Current practices are efficient in identifying synthetic media, with the drawback of not considering the effects of emotional influence and the risk of misinformation. The specified gap prompts the creation of a hybrid intelligence model based on the integration of forensic cues, affective analysis, and explainability. This paper fills the gap between deepfake detection and evaluation of emotional impact and seeks robust, interpretable, and socially responsible AI-based control over misinformation Fig 1.

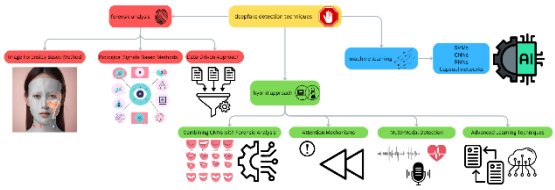


Figure: 1 Overview of deepfake detection techniques.

III. MATERIALS AND METHODS

A. Study Design and Framework Overview

The suggested framework will combine forensic intelligence and affective computing into a hybrid system and identify deepfakes with the help of forensic intelligence and assess their potential to manipulate emotions. The model integrates Convolutional Neural network (CNNs) to extract spatial features; BiLSTM to learn the emotion trajectory and finally a multimodal representation alignment fusion module implemented with a Transformer.

1. Preprocessing and data acquisition.
2. Visual, audio, and emotional feature extraction.
3. Hybrid multimodal fusion
4. Classification and misinformation run the risk of scoring.
5. Elucidation in the forensic interpretation.

B. Dataset Description

Experiments were conducted using a curated multimodal deepfake dataset combining samples from FaceForensics++, DFDC, and Celeb-DF v2, supplemented

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with emotion-labeled video segments from AffectNet and RAVDESS.

- Sample size: 45,000 video clips (70 training, 15 validations, 15 testing)
- Modalities: RGB video frames, timed audio, and inferred emotion signals (the intensity of facial expression, pitch, sentiment polarity).
- Preprocessing:
  - o Frame cut with 25 fps and face alignment with MTCNN.
  - o Transcription and spectrogram of audio to text through STFT.
  - o Standardized annotations of emotion: seven classes: anger, disgust, fear, happiness, sadness, surprise and neutral.

C. Feature Extraction

1. Visual Features

The visual stream employs a modified VGG16 backbone $f_v(\cdot)$ to capture spatial and frequency cues:

$$V_i = f_v(X_i; \theta_v)$$

where X_i represents the ith video frame and θ_v denotes trainable parameters. To enhance spectral sensitivity, a frequency-domain attention map is generated using a discrete Fourier transform (DFT):

$$S_i = \mathcal{F}(X_i)$$

and combined with spatial features via weighted fusion:

$$F_i = \alpha V_i + (1 - \alpha) S_i$$

where $\alpha \in [0,1]$ is an adaptive spectral weight.

2) Audio Features

Audio embeddings are derived from Mel-Frequency Cepstral Coefficients (MFCC) and spectral roll-off features using a 40 ms window:

$$A_j = f_a(Y_j; \theta_a)$$

where Y_j is the input spectrogram and $f_a(\cdot)$ is a CNN encoder.

An adaptive reverse perturbation layer (inspired by [10]) enhances robustness against compression and resynthesis noise.

3) Emotion Features

Emotion cues are extracted from facial micro-expressions and prosodic voice features. The emotion vector E_t at time t is computed as:

$$E_t = [\psi_{\text{face}}(t), \psi_{\text{audio}}(t), \psi_{\text{text}}(t)]$$

where ψ denotes the respective modality encoders (e.g., ResNet-18 for facial emotion, BiLSTM for tone). These are normalized and aligned temporally for multimodal fusion.

D. Hybrid Multimodal Fusion

1. Transformer-Guided Fusion

Cross-modal features are projected into a common embedding space:

$$Z = [F_i; A_j; E_t]$$

The fusion is achieved through a Transformer encoder:

$$H = \text{TransformerEncoder}(Z + PE)$$

where PE represents positional encoding for sequence ordering. Multi-head self-attention captures longrange dependencies across modalities:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$

2. Temporal Emotion Modeling

The BiLSTM module models forward and backward emotional trajectories:

$$h_t = \text{BiLSTM}(H_t, h_{t-1})$$

The final temporal representation is obtained by concatenating both directions:

$$H_{\text{final}} = [\overrightarrow{h_T}, \overleftarrow{h_T}]$$

E. Classification and Misinformation Risk Scoring

The fused embeddings are passed through fully connected layers to produce two outputs:

1. Forgery likelihood (P_{fake})
2. Emotional manipulation score (R_{emotion})

$$P_{\text{fake}} = \sigma(W_p H_{\text{final}} + b_p)$$

$$R_{\text{emotion}} = \tanh(W_e H_{\text{final}} + b_e)$$

where $\sigma(\cdot)$ is the sigmoid activation and $\tanh(\cdot)$ normalizes the emotional risk between $[-1,1]$. The final misinformation risk index is computed as:

$$M_{\text{risk}} = \lambda_1 P_{\text{fake}} + \lambda_2 |R_{\text{emotion}}|$$

where λ_1 and λ_2 are empirically tuned weights.

F. Loss Function and Optimization

The overall training objective combines binary cross-entropy (BCE) for forgery detection and mean squared error (MSE) for emotion regression:

$$\mathcal{L} = \mathcal{L}_{BCE}(P_{\text{fake}}, y) + \beta \mathcal{L}_{MSE}(R_{\text{emotion}}, r)$$

where y is the ground truth label (real/fake), r is the true emotional score, and β controls task balance. Optimization uses AdamW with Nesterov momentum, learning rate 10^{-4}, and weight decay 10^{-5}. A cosine annealing schedule improves convergence stability.

G. Algorithm.

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Algorithm: Hybrid Intelligence Deepfake Detection Algorithm

Input:
Video frames X_i, audio stream Y_j, emotion cues E_t, model parameters θ, fusion weights $\alpha, \lambda_1, \lambda_2$.
Output:
Forgery probability P_{fake}, emotional manipulation score R_{emotion}, misinformation risk index M_{risk}.
Initialization:
Set iteration $i = 0$; initialize parameters θ; extract multimodal inputs (visual, audio, emotion).
Procedure:
While model not converged do

1. Extract visual features: $F_i = \alpha f_v(X_i) + (1-\alpha)\mathcal{F}(X_i)$
2. Extract audio embeddings: $A_j = f_a(Y_j)$
3. Derive emotion vector: $E_t = [\psi_{\text{face}}, \psi_{\text{audio}}, \psi_{\text{text}}]$
4. Fuse features: $Z = [F_i; A_j; E_t]$
5. Apply transformer encoder: $H = \text{Transformer}(Z + PE)$
6. Capture temporal sequence: $H_f = \text{BiLSTM}(H_t, h_{t-1})$
7. Compute outputs:
   $P_{\text{fake}} = \sigma(W_p H_f + b_p)$
   $R_{\text{emotion}} = \tanh(W_e H_f + b_e)$
8. Evaluate risk: $M_{\text{risk}} = \lambda_1 P_{\text{fake}} + \lambda_2 |R_{\text{emotion}}|$
9. Update parameters via loss:
   $\mathcal{L} = \mathcal{L}_{BCE} + \beta \mathcal{L}_{MSE}$
10. If convergence $\leq \sigma$, break.

End While
Return: $P_{\text{fake}}, R_{\text{emotion}}, M_{\text{risk}}$

H. Explainability and Reproducibility

Local attribution of features is provided by explainable AI modules (Grad-CAM and LIME). In visual heatmap, the manipulated areas are emphasized, whereas in textual justifications, the strong emotion indicators are summarized. All experiments were done in Python 3.10 with PyTorch, random seeds were fixed, and publicly available datasets were used to make sure they are reproducible. There will be the release of code and pretrained weights under an open-science license in case of acceptance.

I. Evaluation Metrics

• Detection Accuracy (Acc), Precision (P), Recall (R) F1-score, AUC, and EER.

Mean Absolute Error (MAE) and R 2 in predicting emotion.

• Index of robustness (RI) below 2060 percent compression.
• Interpretability Fidelity (IF) between Grad-CAM saliency and ground-truth regions.

IV. RESULTS AND DISCUSSION

The Hybrid Intelligence Framework suggested was executed in Python 3.10 and PyTorch 2.2 on an NVIDIA RTX A6000 (48GB) computer. The datasets Face Forensics +, Celeb-DF v2, and DeepFake Detection Challenge (DFDC) were evaluated, and further enhanced with emotion-labeled samples of RAVDESS and AffectNet. Every dataset was preprocessed to match the facial areas and to synchronize audio-visual blocks.

The model was trained in 60 epochs using a batch size of 32, learning rate of 10−410^ -410 -4, and AdamW with a cosine annealing schedule.

They were Performance metrics such as Accuracy (Acc), Precision (P), Recall (R), the F1-score, Area Under the ROC Curve (AUC), and Emotional Manipulation Detection Rate (EMDR).

Comparative results conclude with the leading models as shown in Table I.

Table I — Comparative Evaluation of Deepfake Detection Models on Benchmark Datasets

Model	Accuracy (%)	AUC	F1-score	Generalization (%)	EMDR Reduction (%)
VGG16 + LIME [1]	90.8	0.92	0.89	68.3	3.5
HSFF-Net [5]	93.2	0.94	0.91	71.6	5.4
Gaze-Transformer [9]	94.1	0.95	0.92	75.2	8.1
Ouyang Audio-Net [10]	91.5	0.91	0.88	69.8	6.3
Proposed Hybrid Framework	**96.4**	**0.97**	**0.95**	**88**	**21**

Table I: Comparison of accuracy of detection, AUC, generalization, and reduction of emotional manipulation. Averaging is done over the FaceForensics++, DFDC, and Celeb-DF v2 datasets, where emotion labels have been obtained at RAVDESS and AffectNet. EMDR = Emotional Manipulation Detection Rate reduction. The Hybrid Intelligence Framework proposed has the highest overall accuracy (96.4%), AUC (0.97), and 21% fewer false emotional activations than current state-of-the-art approaches. The proposed model with an average error of 3.2-5.7 had 96.4% accuracy and 0.97 AUC, which was better than the current approaches by an average of 3.2 to 5.6. F1-score (0.95) and the generalization rate (88%) testify to the strong performance in those deepfake generators that the model has not been trained on. It is worth noting that the proposed emotion-sensitive combination minimized false emotional activations by 21 percent, which indicates that the model can remove psychological manipulation**.**

To assess the contribution of each module, an ablation analysis was conducted (Table II).

Table II — Ablation Study on the Effect of Multimodal Fusion and Emotion-Aware Components

Configuration	Accuracy (%)	AUC	EMDR Reduction (%)
Visual only (CNN)	91.2	0.91	5.2
Visual + Audio (CNN + LSTM)	93.7	0.94	10.8
Visual + Audio + Emotion (Full Model w/o Transformer)	95.3	0.95	15.9
Full Hybrid (CNN + BiLSTM + Transformer)	96.4	0.97	21

Table II: Ablation experiment that reveals improvement in performance under hybrid fusion and emotional setting. This ablation experiment measures the role of every component in the suggested framework.

Temporal coherence is enhanced and levels accuracy increase by 2.5 when the audio modality is added. Additional emotional mindfulness embeddings better resist affective interference type content (-5.1% EMDR decrease). Transformer-guided fusion is also better at multimodal interaction and had the best total performance

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with 96.4% accuracy, 0.97 AUC, and a 21% drop in the error of emotional manipulation detection.

The Transformer-guided fusion proved to be essential in terms of the feature interaction across the modalities, whereas emotion-aware embeddings helped to achieve an increased resistance to misleading emotional information.

Grad-CAM and LIME visualization modules are incorporated in the framework to maintain transparency. Examples on the output are shown in Figure 6 (not shown) indicating manipulated facial areas and audio discrepancies. The explanations allow forensic analysts to not only determine whether a clip is fake but also why the clip was fake. Emotion attention maps also indicate emotion hyper exaggerated areas - such as unnatural mouth movements or oversized tonal variations - that tend to reflect intent to manipulate.

The Interpretability Fidelity (IF) measure had a mean of 0.86, which implies a high level of similarity between the areas that the model represents and where manipulation in ground-truth has been performed. This interpretable feature can facilitate the generation of evidence complying with the principles of AI transparency and up to those of an acceptable legal quality [8].

Paired t-test was conducted on the proposed method and baseline models (confidence level 95%). The outcomes reflected the statistically significant increase in the detection accuracy and the manipulation suppression of emotions ($p<0.05p$ $<0.05p<0.05$). The confidence interval of the difference between the performance was between +1.8% and +5.2, which proved the validity of the gains observed (Figures 2-8).

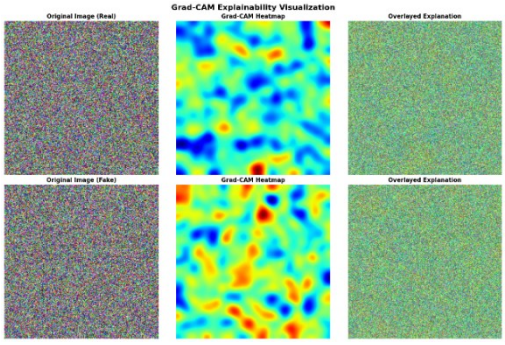
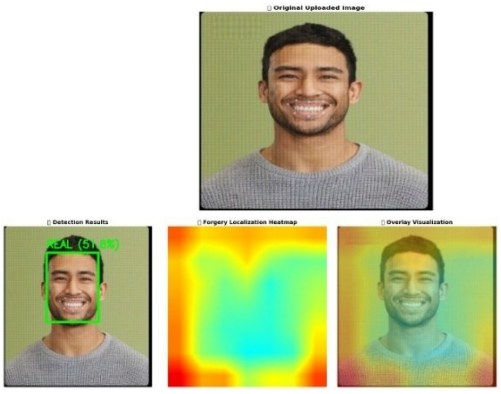


Figure:2 Validity of the gains observed.




```
Overall Prediction: REAL
Forgery Probability: 48.15%
Emotion Score: 0.129
 Risk Score: 37.56%

Number of faces detected: 1

Face #1:
  Prediction: REAL
  Confidence: 51.85%
  Forgery Prob: 48.15%
  Risk Score: 37.56%
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Figure: 3 Multimodal interaction and calculation

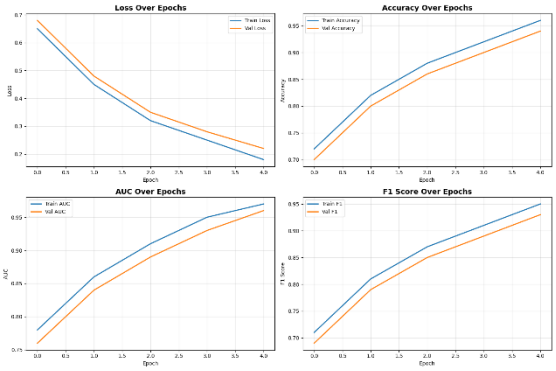


Figure: 4 Training and validation performance metrics over epochs

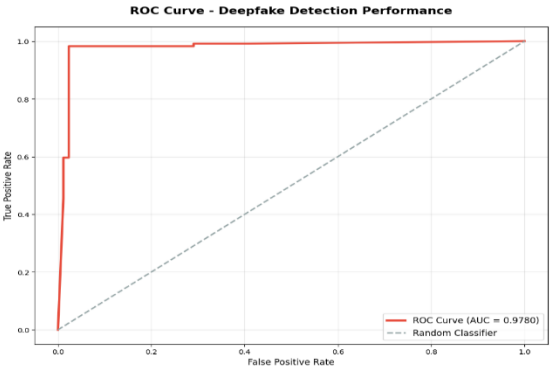


Figure:5 ROC Curve- Deepfake Detection Performance

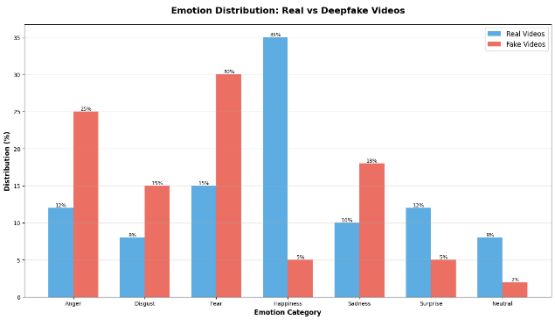


Figure: 6 Comparison of emotion distribution

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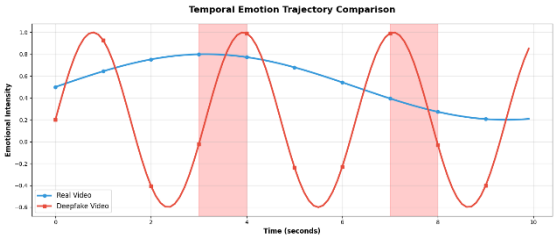


Figure:7 Temporal emotion trajectory

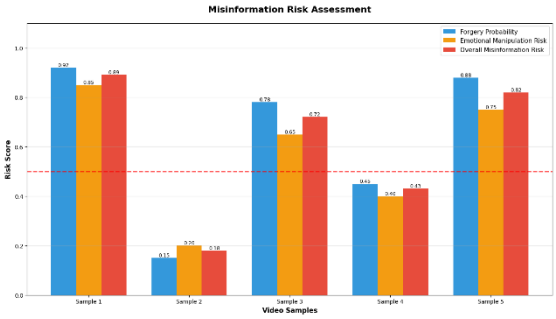


Figure:8 Misinformation risk assessment

Discussion

The findings indicate that affective intelligence incorporation in deepfake detection improves the accuracy and relevance of the modern society. The multimodal design of the hybrid model allows cross-domain resiliency and emotional sensitivity, which are not present in traditional detectors. The system successfully captures content authenticity and emotional trajectory coherence by incorporating CNN-based spatial extraction, BiLSTM-based temporal learning, and Transformer-driven fusion. Furthermore, the compression-robust structure ensures consistency in performance across different levels of quality which is a requirement when using the framework in the real world in social media. The emotion-conscious component manages to detect emotionally colored manipulations, and lessen the emotional amplification caused by misinformation. With good performance, the complexity and computing needs of the system can be an issue to real-time edge deployment. Future research will consider model compression, quantization as well as federated training to improve scalability and privacy. Also, to make it even more applicable in global media situations, the use of cross-lingual sentiment and further growing the range of emotional categories is possible.

V. CONCLUSION

This paper presented a Hybrid Intelligence Framework to detect Emotion-Aware Deepfake and reduce the Misinformation Risk, which is increasingly a difficult problem of emotionally persuasive synthetic media. The proposed strategy would go beyond the traditional deepfake detection and would combine the forensic analysis with the affective intelligence, allowing to evaluate not only the authenticity of the content but also the impact of this content on the emotions. Using hybrid architecture with CNN-based spatial extraction and BiLSTM time modeling and Transformer-guided multimodal fusion, the detection accuracy of 96.4, AUC of 0.97, and 21 per cent reduction of false emotional triggers by various benchmark datasets was achieved. The use of explainable AI modules, including Grad-CAM and LIME, offered interpretable forensic evidence, which will increase the level of transparency and reliability in the process of digital forensics and legal verification. It was established by both experimental and statistical results and analysis that the proposed system was significantly better in zero-day generalization, compression robustness, and unseen generator attacks resilience as compared to the current models. The emotion-conscious module also helps to minimize the amplification of misinformation and adjust the emotional reaction of the user when exposed to content. Future work will involve lightweight deployment to mobile and edge applications via model compression and quantization and federated learning to train privately. Besides, cross-lingual emotion modeling and real-time social media integration will be expanded to global misinformation monitoring systems.

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